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SEMICONDUCTOR STRUCTURE AND MANUFACTURING METHOD THEREOF

Abstract

Provided are a semiconductor structure and a manufacturing method thereof according to embodiments of the present disclosure. The semiconductor structure includes: a carrier plate; light-emitting units on the carrier plate and lenses on a side of the light-emitting units away from the carrier plate. Each of the light-emitting units includes a first light-emitting structure, a second light-emitting structure and a third light-emitting structure, that are spaced apart on a surface of the carrier plate and respectively configured to emit light with different wavelengths, the second light-emitting structure surrounds the first light-emitting structure, and the third light-emitting structure surrounds the second light-emitting structure. The lenses respectively correspond to the light-emitting units, and the lenses are configured to converge light emitted by the light-emitting units and mix the colors of the light emitted by the light-emitting units.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Chinese Patent Application No. 2024102320667 entitled “SEMICONDUCTOR STRUCTURE AND MANUFACTURING METHOD THEREOF” filed on Feb. 29, 2024, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of semiconductors and in particular, to a semiconductor structure and a manufacturing method thereof.

BACKGROUND

[0003] With the development of display technology, light-emitting devices such as Light emitting diodes (LED), Organic light emitting diode (OLED), and Liquid crystal displays (LCD) are widely used in electronic products such as computers, televisions, mobile phones and wearable devices. Micro Light emitting diode (Micro LED) is an emerging technology mainly based on inorganic GaN-based LEDs. Compared Micro LED with LCD and OLED, Micro LED has advantages such as small size, high contrast ratio, low power consumption and long service life, etc. However, the current display effect of Micro LED display panels still needs to be improved, and the manufacturing process of Micro LED display panels is relatively complex, thereby resulting in high production cost.

SUMMARY

[0004] In view of this, a semiconductor structure and a manufacturing method thereof are provided in the present disclosure.

[0005] In a first aspect, a semiconductor structure is provided according to the present disclosure. The semiconductor structure includes: a carrier plate; light-emitting units on the carrier plate and lenses on a side of the light-emitting units away from the carrier plate. Each of the light-emitting units includes a first light-emitting structure, a second light-emitting structure and a third light-emitting structure that are spaced apart on a surface of the carrier plate and respectively configured to emit light with different wavelengths, the second light-emitting structure surrounds the first light-emitting structure, and the third light-emitting structure surrounds the second light-emitting structure. The lenses respectively correspond to the light-emitting units, and the lenses are configured to converge light emitted by the light-emitting units.

[0006] In a second aspect, a manufacturing method for a semiconductor structure is provided according to the present disclosure. The method includes: forming light-emitting units, where each of the light-emitting units comprises a first light-emitting structure, a second light-emitting structure and a third light-emitting structure that are spaced apart and respectively configured to emit light with different wavelengths, the second light-emitting structure surrounds the first light-emitting structure, and the third light-emitting structure surrounds the second light-emitting structure; bonding the light-emitting units to a carrier plate; and forming lenses on a side of the light-emitting units away from the carrier plate, where the lenses respectively correspond to the light-emitting units, and the lenses are configured to converge light emitted by the light-emitting units.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] FIG. **1** is a schematic diagram of a top view for a semiconductor structure according to an embodiment of the present disclosure.

[0008] FIG. **2** is a schematic diagram of a section along line AB in FIG. **1**.

[0009] FIG. **3** is a schematic diagram of a section for a semiconductor structure according to an embodiment of the present disclosure.

[0010] FIG. **4** is a schematic diagram of a top view for a light-emitting unit according to an embodiment of the present disclosure.

[0011] FIG. **5** is a schematic diagram of a top view for a light-emitting unit according to another embodiment of the present disclosure.

[0012] FIG. **6** is a schematic diagram of a top view for a light-emitting unit according to a still another embodiment of the present disclosure.

[0013] FIG. **7** is a schematic diagram of a localized section for another semiconductor structure according to an embodiment of the present disclosure.

[0014] FIG. **8** is a schematic diagram of a localized section for another semiconductor structure according to an embodiment of the present disclosure.

[0015] FIG. **9** is a schematic diagram of a localized section for another semiconductor structure according to an embodiment of the present disclosure.

[0016] FIGS. **10** to **18** are schematic diagrams of localized sections for intermediate structures of a light-emitting unit formed by a manufacturing method of a semiconductor structure according to an embodiment of the present disclosure.

[0017] FIGS. **19** to **21** are schematic diagrams of localized sections for intermediate structures of a light-emitting unit formed by another manufacturing method of a semiconductor structure according to an embodiment of the present disclosure.

[0018] FIGS. **22** and **23** are schematic diagrams of localized sections for intermediate structures formed by another manufacturing method of a semiconductor structure according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0019] To enable those in the art to better understand the solution of the present disclosure, the technical solutions in embodiments of the present disclosure will be clearly and completely described below in conjunction with the accompanying drawings in embodiments of the present disclosure. Obviously, the described embodiments are part of embodiments of the present disclosure, but not all of the embodiments. It should be understood that the terms first, second, etc. used in the present disclosure are only used to distinguish the same type of information from each other, but are not necessarily used to describe a specific order or sequence.

[0020] A semiconductor structure is provided according to the present disclosure, and this semiconductor structure can mitigate the problem of low luminous intensity of Micro LED, as well as high production cost.

[0021] FIG. **1** is a schematic diagram of a top view for a semiconductor structure according to an embodiment of the present disclosure. FIG. **2** is a schematic diagram of a section along line AB in FIG. **1**. As shown in FIGS. **1** and **2**, a semiconductor structure is provided according to an embodiment of the present disclosure, which includes a carrier plate **10**, light-emitting units **20** and lenses **30**.

[0022] In some embodiments, the carrier plate **10** may include a silicon substrate and a driving circuit layer for driving the light-emitting unit **20**.

[0023] In some embodiments, the light-emitting units **20** are on the carrier plate **10**. As shown in FIG. **1**, the light-emitting units **20** are arranged in an array. Each light-emitting unit **20** includes a first light-emitting structure **21**, a second light-emitting structure **22** and a third light-emitting structure **23**, that are spaced apart on a surface of the carrier plate **10** and respectively configured to

emit light with different wavelengths, the second light-emitting structure **22** surrounds the first light-emitting structure **21**, and the third light-emitting structure **23** surrounds the second light-emitting structure **22**.

[0024] In embodiments of the present disclosure, the second light-emitting structure **22** surrounds the first light-emitting structure **21**, and the third light-emitting structure **23** surrounds the second light-emitting structure **22**, and this arrangement manner can make the distribution of the light with different wavelengths emitted by the first light-emitting structure, the second light-emitting structure and the third light-emitting structure uniform in all directions, and can reduce crosstalk between light with different colors. That is, the distribution of the light on a light-emitting surface of the light-emitting unit **20** is uniform. In addition, the second light-emitting structure **22** and the third light-emitting structure **23** are annular structures, that can help to release stress of the second light-emitting structure **22** and the third light-emitting structure **23** and improve the performance of the light-emitting unit **20**.

[0025] In some embodiments, each of the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** may include an N-type semiconductor layer (not shown), an active layer (not shown) and a P-type semiconductor layer (not shown) that are stacked sequentially. Materials of the N-type semiconductor layer may include N-type doped Group III nitride-based materials. The N-type doping element may include at least one of Si, Ge, Sn, Se, or Te. The active layer **22** may include at least one of a single quantum well structure, a multiple quantum well structure, a quantum line structure or a quantum dot structure. Materials of the P-type semiconductor layer may include P-type doped Group III nitride-based materials. The P-type dopant element may be at least one of Mg, Zn, Ca, Sr, or Ba. The group III nitride material may include any one or any combination of GaN, AlGa_N, InGa_N and AlInGa_N.

[0026] In some embodiments, the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** are spaced apart from each other. For example, space between the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** are filled with a dielectric material, for example, the dielectric material may be silicon dioxide.

[0027] In some embodiments, the semiconductor structure may be a Micro Led display panel, and the light-emitting units **20** may be used as pixels that are arranged in an array in the display panel. The first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** may serve as sub-pixels of the display panel. The first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** may respectively emit blue light, green light, and red light.

[0028] The lenses **30** are on a side of the light-emitting units **20** away from the carrier plate **10**, where the lenses **30** respectively correspond to the light-emitting units **20**, and the lenses **30** converge light emitted by the light-emitting units **20**. Specifically, a lens **30** is correspondingly disposed on a light-emitting side of a light-emitting unit **20**, and the lens **30** can converge the light with different wavelengths emitted by the light-emitting unit **20**, thereby achieving the purpose of color mixing and preventing the light emitted by different light-emitting structures from the crosstalk before the light reaching a receiver, thereby avoiding the degradation of the light output effect. In some embodiments, when the semiconductor structure is used for display, the receiver can be the human eye. The light emitted by the semiconductor structure is the light after color mixing to avoid the degradation of the display effect due to the crosstalk of the light. On the other hand, it can be avoided that the human eye recognizes the boundaries of light-emitting structures emitted light with different wavelengths, thereby improving a display experience of a viewer.

[0029] In some embodiments, multiple light-emitting structures with different light-emitting wavelengths of the light-emitting unit **20** can be epitaxially manufactured at the same time to avoid the complicated process of massive transfer of RGB colors in the later stage. Therefore, the process difficulty is reduced, and the yield of the semiconductor structure is improved. In an embodiment,

an orthographic projection of the light-emitting unit **20** on the carrier plate **10** is within the orthographic projection of the corresponding lens **30** on the carrier plate **10**. That is, an area of the orthographic projection of the lens **30** on the carrier plate **10** is greater than or equal to an area of the orthogonal projection of the light-emitting unit **20** on the carrier plate **10**. When the area of the orthographic projection of the lens **30** on the carrier plate **10** and the area of the orthographic projection of the light-emitting unit **20** on the carrier plate **10** are equal, the lens **30** can converge the parallel light emitted from the light-emitting unit **20** to achieve color mixing. When the area of the orthographic projection of the lens **30** on the carrier plate **10** is larger than the area of the orthographic projection of the light-emitting unit **20** on the carrier plate **10**, the lens **30** further can converge the light emitted obliquely outward from the light-emitting unit **20**, which can enhance the intensity of the light emitted from the semiconductor structure in the embodiments of the present disclosure, as well as the luminous efficiency.

[0030] In some embodiments, the lens **30** includes a Fresnel lens. The Fresnel lens is a special lens with a zigzag structure that can converge scattered light with a large viewing angle into parallel light and can control the light in the viewing angle. If the light projected to a side of the Fresnel lens is parallel light, the light emitted out from the other side of the Fresnel lens is converged into one point. Specifically, as shown in FIG. 2, a light-emitting unit **20** faces a lens **30**. The light-emitting unit **20** emits light of multiple wavelengths in parallel. After the light passing through the lens **30**, the light converges and focuses into a point to complete color mixing, thereby improving the display effect.

[0031] In some embodiments, a material of the lens **30** may be a transparent material. For example, the material of the lens **30** may include a material with high light transmittance such as organic polymer, glass or etc. where the organic polymer may include polyolefins.

[0032] In some embodiments, the material of the lens **30** includes SiO₂, GaN or AlN. The lens **30** is manufactured on the light-emitting unit **20** through a semiconductor process. The process is simple and the light-emitting unit **20** and the lens **30** can be integrated at one time.

[0033] In some embodiments, as shown in FIG. 2, the light-emitting unit **20** and the lens **30** are spaced apart from each other by a certain distance.

[0034] In some embodiments, FIG. 3 is a schematic diagram of a section for a semiconductor structure according to an embodiment of the present disclosure. As shown in FIG. 3, light-emitting units **20** are disposed on the carrier plate **10**, and the light from each light-emitting unit **20** passes through a lens **30** to converge at an imaging position **301** and is subsequently received by the receiver (such as the human eye). Specifically, the imaging position **301** may be a physical structure, or may refer to the convergence position of the light.

[0035] In an embodiment, a center of gravity of the orthographic projection of the light-emitting unit **20** on the carrier plate **10** coincides with a center of gravity of the orthographic projection of the lens **30** on the carrier plate **10**, which can improve the color mixing effect of the lens **30**. In some embodiments, the center of gravity of the orthographic projection of the first light-emitting structure **21** on the carrier plate **10**, the center of gravity of the orthographic projection of the second light-emitting structure **22** on the carrier plate **10**, and the center of gravity of the orthographic projection of the third light-emitting structure **23** on the carrier plate **10** coincide with each other, which is convenient for the lens **30** to mix colors. In some embodiments, the center of gravity of the orthographic projection of the first light-emitting structure **21** on the carrier plate **10**, the center of gravity of the orthographic projection of the second light-emitting structure **22** on the carrier plate **10**, and the center of gravity of the orthographic projection of the third light-emitting structure **23** on the carrier plate **10** further coincide with the center of gravity of the orthographic projection of the lens **30** on the carrier plate **10**, which further improve the color mixing effect of the lens **30**.

[0036] In an embodiment, the orthographic projections of the second light-emitting structure **22** and the third light-emitting structure **23** on the carrier plate **10** are respectively annulus, and the

contour shape of the annulus is the same as the contour shape of the first light-emitting structure **21**. In an embodiment, as shown in FIG. **1**, the orthographic projection of the first light-emitting structure **21** on the carrier plate **10** is a circle, and the orthographic projections of the second light-emitting structure **22** and the third light-emitting structure **23** on the carrier plate **10** are circular annuluses respectively, that is, the shape of the light-emitting structure is the circular or the circular annulus, which helps to release stress and avoid cracking during the process of manufacturing or using. In another embodiment, FIG. **4** is a schematic diagram of a top view for a light-emitting unit according to an embodiment of the present disclosure. As shown in FIG. **4**, the orthographic projection of the first light-emitting structure **21** on the carrier plate **10** is a rectangle, and the orthographic projection of the light-emitting structure **22** and the third light-emitting structure **23** on the carrier plate **10** are respectively rectangular annuluses. In another embodiment, FIG. **5** is a schematic diagram of a top view for a light-emitting unit according to an embodiment of the present disclosure. As shown in FIG. **5**, the orthographic projection of the first light-emitting structure **21** on the carrier plate **10** is ellipsoidal, and the orthographic projection of the light-emitting structure **22** and the third light-emitting structure **23** on the carrier plate **10** are respectively ellipsoidal annuluses. Specifically, the light emitted by the light-emitting unit **20** is uniformly emitted out from the center of the light-emitting unit **20** along a radiation direction of the contour shape. Therefore, the display device manufactured by the semiconductor structure according to the embodiments of the present disclosure can have an excellent display effect.

[0037] In some embodiments, the orthographic projection of the first light-emitting structure **21** on the carrier plate **10** is circular, polygonal, ellipsoidal, or other special-shaped.

[0038] As shown in FIG. **1**, in an embodiment, a shape of the orthographic projection of the lens **30** on the carrier plate **10** is the same as a shape of the orthographic projection of the light-emitting unit **20** on the carrier plate **10**. For example, the light-emitting unit **20** is circular, and the lens **30** is also circular; the light-emitting unit **20** is rectangular, and the lens **30** is also rectangular. The shape of the lens **30** and the shape of the light-emitting unit **20** are the same may result in that the edge(s) of the lens **30** and the edge(s) of the light-emitting unit are parallel, and the parallel light is converged through the lens **30**, so the light emitted by the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure can be better converged. In some embodiments, the area of the orthographic projection of the lens **30** on the carrier plate **10** is greater than or equal to the area of the orthographic projection of the light-emitting unit **20** on the carrier plate **10**.

[0039] In an embodiment, FIG. **6** is a schematic diagram of a top view for another light-emitting unit according to an embodiment of the present disclosure. As shown in FIG. **6**, the second light-emitting structure **22** includes at least two second sub-light-emitting structures which are spaced apart; and/or the third light-emitting structure **23** includes at least two third sub-light-emitting structures which are spaced apart. In some embodiments, the number of the second sub-light-emitting structures may be 2, 3, 4, etc.; or the number of the third sub-light-emitting structures may be 2, 3, 4, etc. As shown in FIG. **6**, the second light-emitting substructure **22** includes four second sub-light-emitting structures, and the third light-emitting structure **23** includes eight third sub-light-emitting structures. In some embodiments, the second sub-light-emitting structures can be controlled separately, and the third sub-light-emitting structures can be controlled separately, that may result in that the luminous intensity of the light-emitting structure can be adjusted more flexibly.

[0040] In an embodiment, the semiconductor structure includes a first electrode layer **40** and a second electrode layer **50**.

[0041] FIG. **7** is a schematic diagram of a localized section for another semiconductor structure according to an embodiment of the present disclosure. As shown in FIG. **7**, in some embodiments, the first electrode layer **40** is between the light-emitting unit **20** and the carrier plate **10** and includes first electrodes **41** that are electrically connected to the first light-emitting structure **21**, the

second light-emitting structure **22** and the third light-emitting structure **23** respectively. The second electrode layer **50** is on the side of the light-emitting unit **20** away from the carrier plate **10** and includes three second electrodes **51** that are electrically connected to the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** respectively. The second electrodes **51** are electrically isolated from each other. That is, the first electrode **41** is an electrode shared by the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23**, so that it is possible to form the first electrode **41** connected to the first light-emitting structure **21**, the second light-emitting structure **22**, and the third light-emitting structure **23** in a single process, and thus efficiency can be improved. At this time, the carrier plate **10** is not the growth substrate of the light-emitting unit **20**.

[0042] In some embodiments, the first electrode **41** shared by the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** may be disposed on the side of the light-emitting unit **20** away from the carrier plate **10**. Three second electrodes **51** respectively connected to the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** are disposed at a side of the light-emitting unit **20** facing the carrier plate **10**.

[0043] In some embodiments, the materials of the first electrode **41** and the second electrode **51** may be metal materials respectively, for example, the materials of the first electrode **41** and the second electrode **51** may be copper, silver, iron, and alloys thereof respectively.

[0044] FIG. **8** is a schematic diagram of a localized section for another semiconductor structure according to an embodiment of the present disclosure. In some embodiments, as shown in FIG. **8**, the first electrode layer **40** is between the light-emitting unit **20** and the carrier plate **10** and includes third electrodes **43** and fourth electrodes **44** which are respectively electrically connected to the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23**. The first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** all include two semiconductor layers with different conductivity types, the third electrode **43** and the fourth electrode **44** are electrically connected to the two semiconductor layers respectively (e.g. the third electrode **43** is electrically connected to the semiconductor layer with a conductivity type and the fourth electrode **44** is electrically connected to the semiconductor layer with another conductivity type), so that the light emitted by the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** can be controlled separately.

[0045] In some embodiments, the electrical signals of the first electrode layer **40** and the second electrode layer **50** are controlled to achieve full color or white light display. Specifically, taking the semiconductor structure shown in FIG. **7** as an example, a first common signal is provided for the first electrode layer **40** and different second signals are provided for the second electrodes **51** of the first light-emitting structures **21**, the second light-emitting structures **22** and the third light-emitting structures **23** in the light-emitting units **20**. After the light is mixed by the lens **30**, the light-emitting units **20** emit light with different wavelengths ranges and/or different intensity, thereby achieving full color display ultimately. Specifically, taking the semiconductor structure shown in FIG. **7** as an example, a first common signal is provided for the first electrode layer **40** and the same second signals are provided for the second electrodes **51** of the first light-emitting structures **21**, the second light-emitting structures **22** and the third light-emitting structures **23** in the light-emitting units **20**. After the light is mixed by the lens **30**, the light-emitting units **20** emit white light, thereby achieving white display ultimately.

[0046] In an embodiment, a light-emitting structure among the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** emitted light with the largest wavelength has a largest projection area on the carrier plate **10**. For example, when the luminous colors of the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** are blue, green, and red respectively, the projected areas of the

first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** on the carrier plate **10** increase sequentially. Since the luminous efficiency of light with different wavelengths is different, the shorter the wavelength, the better the luminous efficiency. An increased area of the light-emitting structure emitted light with a larger emission wavelength may increase the intensity of the light emitted by this light-emitting structure. In addition, the human eye has different sensitivity to light with different wavelengths. Based on above reasons, by designing the area ratios of different light-emitting structures, the intensity of light emitted by the light-emitting structures in the light-emitting unit **20** can be effectively balanced and the problem of a color shift of the display can be mitigated.

[0047] In some embodiments, the semiconductor structure provided by this disclosure is used to make a vehicle light array.

[0048] FIG. **9** is a schematic diagram of a localized section for another semiconductor structure according to an embodiment of the present disclosure. In an embodiment, as shown in FIG. **9**, each of the light-emitting structures of the light-emitting unit **20** includes a light-emitting layer **200** and a light conversion layer **201** on the light-emitting layer **200**, and the light conversion layer **201** is between the light-emitting layer **200** and the lens **30**. The light conversion layers **201** of the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** are different, so that the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** emitted light with different wavelengths. Specifically, the light-emitting layers **200** of each light-emitting unit **20** emit light with the same wavelength, and then emit out light with different wavelengths after passing through the different light conversion layers **201** of the three light-emitting structures **21/22/23**.

[0049] In some embodiments, the material of the light conversion layer **201** may be quantum dot or phosphor. Specifically, the light-emitting layer **200** emits white light, yellow light or blue light, the light conversion layer **201** in the first light-emitting structure **21** is provided with blue light quantum dots, and finally emits out blue light; the light conversion layer **201** in the second light-emitting structure **22** is provided with green light quantum dots, and finally emits out green light; and the light conversion layer **201** in the third light-emitting structure **23** is provided with red light quantum dots, and finally emits out red light.

[0050] In an embodiment, a manufacturing method for a semiconductor structure is provided according to the present disclosure, which includes the following steps.

[0051] Step **S1**, light-emitting units **20** is formed. Each light-emitting unit **20** includes a first light-emitting structure **21**, a second light-emitting structure **22** and a third light-emitting structure **23** that are spaced apart and respectively configured to emit light with different wavelengths. For each light-emitting unit, the second light-emitting structure **22** surrounds the first light-emitting structure **21**, and the third light-emitting structure **23** surrounds the second light-emitting structure **22**.

[0052] Step **S2**, the light-emitting units **20** and the carrier plate **10** are bonded.

[0053] Step **S3**, lenses **30** are formed on a side of the light-emitting units **20** away from the carrier plate **10**. The lenses **30** respectively correspond to the light-emitting units **20**. The lenses **30** are configured to converge the light emitted by the light-emitting units **20**.

[0054] FIGS. **10** to **18** are schematic diagrams of localized sections for intermediate structures of a light-emitting unit formed by a manufacturing method of a semiconductor structure according to an embodiment of the present disclosure. As shown in FIGS. **10** to **18**, forming the light-emitting unit **20** includes the following steps.

[0055] Step **1101**, a growth substrate **60** is provided.

[0056] As shown in FIG. **10**, the growth substrate **60** may include sapphire, silicon carbide, single crystal silicon, polycrystalline silicon, diamond, gallium nitride, or composite substrate, etc.

[0057] Step **1102**: a first dielectric layer **71** covering the growth substrate **60** is formed. The first dielectric layer **71** includes at least one first groove **81** exposing the growth substrate **60**.

[0058] As shown in FIG. **11**, the first groove **81** may be formed by etching the first dielectric layer

71 using a mask to expose the growth substrate **60**. The shape of an orthographic projection of the first groove **81** on the growth substrate **60** may be circular, polygonal, or ellipsoidal, etc. For example, the shape of the orthographic projection of the first groove **81** on the growth substrate **60** is circular.

[0059] Step **1103**, the first light-emitting structure **21** is epitaxially manufactured in the first groove **81**.

[0060] As shown in FIG. **12**, the first light-emitting structure **21** may include an N-type semiconductor layer (not shown), an active layer (not shown), and a P-type semiconductor layer (not shown) that are stacked in sequence. In some embodiments, the first light-emitting structure **21** may further include a transparent electrode (not shown). For example, the material of the transparent electrode may be ITO (Indium tin oxide). For example, the first light-emitting structure **21** is used for emitting blue light.

[0061] Step **1104**: a second dielectric layer **72** covering the first light-emitting structure **21** and the first dielectric layer **71** is formed.

[0062] As shown in FIG. **13**, the second dielectric layer **72** covers the first light-emitting structure **21** to protect the first light-emitting structure **21** and prevent layers of the second light-emitting structure **22** from being formed on the first light-emitting structure **21** in the subsequent process.

[0063] Step **1105**, the second dielectric layer **72** is etched to form at least one second groove **82** surrounding the first light-emitting structure **21** and exposing the growth substrate **60**.

[0064] As shown in FIG. **14**, the second groove **82** may be formed by etching the second dielectric layer **72** and the first dielectric layer **71** using a mask to expose the growth substrate **60**. For example, the shape of the orthographic projection of the second groove **82** on the growth substrate **60** may be a circular annulus.

[0065] Step **1106**, the second light-emitting structure **22** is epitaxially manufactured in the second groove **82**.

[0066] As shown in FIG. **15**, the layers of the second light-emitting structure **22** are similar with the layers of the first light-emitting structure **21**, and the difference between the layers of the second light-emitting structure **22** and the layers of the first light-emitting structure **21** is that the material of the active layer of the second light-emitting structure **22** is different from the material of the active layer of the first light-emitting structure **21**. In some embodiments, the second light-emitting structure **22** is used for emitting green light.

[0067] Step **1107**: a third dielectric layer **73** covering the second light-emitting structure **22** and the second dielectric layer **72** is formed.

[0068] As shown in FIG. **16**, the third dielectric layer **73** covers the second light-emitting structure **22** and the second dielectric layer **72**. The third dielectric layer **73** is used to prevent the layers of the third light-emitting structure **23** from being formed on the second light-emitting structure **22** in the subsequent process.

[0069] Step **1108**: the third dielectric layer **73** is etched to form at least one third groove **83** surrounding the second light-emitting structure **22** and exposing the growth substrate **60**.

[0070] As shown in FIG. **17**, the third groove **83** may be formed by etching the third dielectric layer **73**, the second dielectric layer **72** and the first dielectric layer **71** using a mask to expose the growth substrate **60**. For example, the shape of the orthographic projection of the third groove **83** on the growth substrate **60** may be a circular annulus.

[0071] Step **1109**, the third light-emitting structure **23** is epitaxially manufactured in the third groove **83**.

[0072] As shown in FIG. **18**, the layers of the third light-emitting structure **23** are similar with the layers of the first light-emitting structure **21**, and the difference between the layers of the third light-emitting structure **23** and the layers of the first light-emitting structure **21** is that the material of the active layer of the third light-emitting structure **23** is different from the material of the active layer of the first light-emitting structure **21**. In some embodiments, the third light-emitting structure

23 is used for emitting red light.

[0073] In some embodiments, after the third light-emitting structure **23** is formed, a planarization process for the first dielectric layer **71**, the second dielectric layer **72**, and the third dielectric layer **73** is performed to expose the first light-emitting structure **21** and the second light-emitting structure **22**.

[0074] In some embodiments, after the planarization process, a first electrode layer is formed on the surfaces of the first dielectric layer **71**, the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23**. The first electrode layer may include a first electrode shared by the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23**.

[0075] In this embodiment, the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** that emit light with different colors are formed on one growth substrate **60** using an epitaxial process. By repeating steps **1101-1109**, the light emitting units **20** can be obtained. The light-emitting units may be entirely bonded to the carrier plate. This method can simplify the manufacturing process of Micro LED, so the production cost of Micro LED is significantly reduced.

[0076] FIGS. **19** to **21** are schematic diagrams of localized sections for intermediate structures of a light-emitting unit formed by another manufacturing method of a semiconductor structure according to an embodiment of the present disclosure. As shown in FIGS. **19** to **21**, forming the light-emitting unit **20** includes the following steps.

[0077] Step **1201**, a growth substrate **60** is provided. In some embodiments, refer to FIG. **10** and step **S1101**, which will not be repeated here.

[0078] Step **S1202**, as shown in FIG. **19**, a fourth dielectric layer **7** covering the growth substrate **60** is formed.

[0079] Step **S1203**, the fourth dielectric layer **74** is etched to form at least one fourth groove **84**, at least one fifth groove **85**, and at least one sixth groove **86** that expose the growth substrate **60** respectively. The fifth groove **85** surrounds the fourth groove **84**, and the sixth groove **86** surrounds the fifth groove **85**. The projected areas of the fourth groove **84**, the fifth groove **85** and the sixth groove **86** on the growth substrate **60** are different.

[0080] As shown in FIG. **20**, the fourth dielectric layer may be etched using a mask. In some embodiments, the orthographic projection of the fourth groove **84** on the growth substrate **60** is a circle, and the orthographic projections of the fifth groove **85** and the sixth groove **86** on the growth substrate **60** may be circular annulus.

[0081] Step **S1204**: the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** emitting light with different wavelengths are respectively manufactured in the fourth groove **84**, the fifth groove **85** and the sixth groove **86** in one epitaxial process.

[0082] As shown in FIG. **21**, the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** emitting light with different wavelengths may be manufactured in one epitaxial process using the same process parameters. Since the sizes of the fourth groove **84**, the fifth groove **85** and the sixth groove **86** are different, the materials of the active layers formed in different grooves are different under the same conditions, the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** formed in one epitaxial process respectively emit blue light, green light, and red light.

[0083] As shown in FIG. **22**, in step **S2**, bonding the light-emitting unit **20** to the carrier plate **10** may include bonding a side of the light-emitting unit **20** away from the growth substrate **60** to the carrier plate **10**.

[0084] As shown in FIG. **23**, after the light-emitting unit **20** and the carrier plate **10** are bonded, the original growth substrate **60** is peeled off.

[0085] In some embodiments, after peeling off the original growth substrate **60**, a second electrode

layer is formed on the side of the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** away from the carrier plate **10**, and the second electrode layer includes three second electrodes that are electrically connected to the first light-emitting structure **21**, the second light-emitting structure **22** and the third light-emitting structure **23** respectively.

[0086] As shown in FIG. 2, in step S3, a lens **30** is formed on the side of the light-emitting unit **20** away from the carrier plate **10**. There is a distance between the light-emitting unit **20** and the lens **30**, and the space between the light-emitting unit **20** and the lens **30** may be used to dispose electrodes and other structures. In some embodiments, when the material of the lens **30** is SiO₂, GaN or AlN, the lens may be manufactured through deposition process or epitaxial process on the surface of the light-emitting unit **20**, this process is simple, and the light-emitting unit **20** and the lens **30** can be integrated together in this process. When the material of the lens **30** is organic polymer or glass, the light-emitting unit **20** and the lens **30** can be integrated together through bonding process.

[0087] A manufacturing method for a semiconductor structure is provided according to the embodiments of the present disclosure. The first light-emitting structure, the second light-emitting structure and the third light-emitting structure emitted light with different colors are formed on one growth substrate and can be entirely bonded to the carrier plate at the same time.

[0088] Compared with the traditional mass transfer process in which the light-emitting structures emitted light with different colors are formed on different growth substrates and then the light-emitting structures emitted light with different colors are transferred to a carrier plate, the manufacturing method for the semiconductor structure according to the embodiments of the present disclosure simplify the manufacturing process of Micro LED, and the production cost of Micro LED is significantly reduced. In the semiconductor structure manufactured by this manufacturing method, the light-emitting unit includes a first light-emitting structure, a second light-emitting structure and a third light-emitting structure. The second light-emitting structure surrounds the first light-emitting structure, and the third light-emitting structure surrounds the second light-emitting structure. The lenses respectively correspond to the light-emitting units, and the lenses are configured to converge light emitted by the light-emitting units and mix the colors of the light emitted by the light-emitting units to avoid crosstalk between the light emitted by different light-emitting structures before the light reaching a receiver, thereby avoiding a reduction of the light-emitting effect.

[0089] A semiconductor structure and a manufacturing method thereof are provided according to the embodiments of the present disclosure. The semiconductor structure includes: a carrier plate; light-emitting units on the carrier plate and lenses on a side of the light-emitting units away from the carrier plate. Each of the light-emitting units includes a first light-emitting structure, a second light-emitting structure and a third light-emitting structure, that are spaced apart on a surface of the carrier plate and respectively configured to emit light with different wavelengths, the second light-emitting structure surrounds the first light-emitting structure, and the third light-emitting structure surrounds the second light-emitting structure. The lenses respectively correspond to the light-emitting units, and the lenses converge light emitted by the light-emitting units and mix the colors of the light emitted by the light-emitting units to avoid crosstalk between the light emitted by different light-emitting structures before reaching a receiver, thereby avoiding a reduction of the light-emitting effect.

[0090] The foregoing are only some embodiments of the present disclosure and are not intended to limit the present disclosure, and any modifications, equivalent substitutions, improvements, etc., made within the spirit and principles of the present disclosure shall be included within the scope of protection of the present disclosure.

Claims

1. A semiconductor structure, comprising: a carrier plate; light-emitting units on the carrier plate, wherein each of the light-emitting units comprises a first light-emitting structure, a second light-emitting structure and a third light-emitting structure that are spaced apart on a surface of the carrier plate and respectively configured to emit light with different wavelengths, the second light-emitting structure surrounds the first light-emitting structure, and the third light-emitting structure surrounds the second light-emitting structure; and lenses on a side of the light-emitting units away from the carrier plate, wherein the lenses respectively correspond to the light-emitting units, and the lenses are configured to converge light emitted by the light-emitting units.
2. The semiconductor structure according to claim 1, wherein an orthographic projection of the light-emitting unit on the carrier plate is within an orthographic projection of the corresponding lens on the carrier plate.
3. The semiconductor structure according to claim 2, wherein the lenses comprise Fresnel lens.
4. The semiconductor structure according to claim 3, wherein a material of the lenses comprises one of SiO₂, GaN, AlN, glass or organic polymer material.
5. The semiconductor structure according to claim 2, wherein a center of gravity of the orthographic projection of the light-emitting unit on the carrier plate coincides with a center of gravity of the orthographic projection of the corresponding lens on the carrier plate.
6. The semiconductor structure according to claim 1, wherein orthographic projections of the second light-emitting structure and the third light-emitting structure on the carrier plate are respectively annuluses, and contour shapes of the annuluses are same as a contour shape of the first light-emitting structure.
7. The semiconductor structure according to claim 6, wherein a shape of an orthographic projection of the lens on the carrier plate is same as a shape of an orthographic projection of the light-emitting unit on the carrier plate.
8. The semiconductor structure according to claim 6, wherein the second light-emitting structure further comprises at least two second sub-light-emitting that are spaced apart.
9. The semiconductor structure according to claim 6, wherein the third light-emitting structure further comprises at least two third sub-light-emitting that are spaced apart.
10. The semiconductor structure according to claim 1, wherein the semiconductor structure further comprises: a first electrode layer between the light-emitting unit and the carrier plate, comprising a first electrode electrically connected to the first light-emitting structure, the second light-emitting structure and the third light-emitting structure respectively; a second electrode layer on the side of the light-emitting unit away from the carrier plate, comprising three second electrodes electrically connected to the first light-emitting structure, the second light-emitting structure and the third light-emitting structure respectively, wherein the three second electrodes are electrically isolated from each other.
11. The semiconductor structure according to claim 1, wherein a light-emitting structure among the first light-emitting structure, the second light-emitting structure and the third light-emitting structure emitted light with the largest wavelength has the largest projection area on the carrier plate.
12. The semiconductor structure according to claim 1, wherein each of the light-emitting structures comprises a light-emitting layer and a light conversion layer on the light-emitting layer, and the light conversion layer is between the light-emitting layer and the lens; and the light conversion layers of the first light-emitting structure, the second light-emitting structure and the third light-emitting structure are different to make the first light-emitting structure, the second light-emitting structure and the third light-emitting structure emit light with different wavelengths.
13. A manufacturing method of a semiconductor structure, comprising: forming light-emitting

units, wherein each of the light-emitting units comprises a first light-emitting structure, a second light-emitting structure and a third light-emitting structure that are spaced apart and respectively configured to emit light with different wavelengths, the second light-emitting structure surrounds the first light-emitting structure, and the third light-emitting structure surrounds the second light-emitting structure; bonding the light-emitting units to a carrier plate; and forming lenses on a side of the light-emitting units away from the carrier plate, wherein the lenses respectively correspond to the light-emitting units, and the lenses converge light emitted by the light-emitting units.

14. The method according to claim 13, wherein forming the light-emitting unit comprises: providing a growth substrate; forming a first dielectric layer covering the growth substrate, wherein the first dielectric layer comprises at least one first groove exposing the growth substrate; epitaxially manufacturing a first light-emitting structure in the at least one first groove; forming a second dielectric layer covering the first light-emitting structure and the first dielectric layer; etching the second dielectric layer to form at least one second groove surrounding the first light-emitting structure and exposing the growth substrate; epitaxially manufacturing a second light-emitting structure in the at least one second groove; forming a third dielectric layer covering the second light-emitting structure and the second dielectric layer; etching the third dielectric layer to form at least one third groove surrounding the second light-emitting structure and exposing the growth substrate; and epitaxially manufacturing a third light-emitting structure in the at least one third groove.

15. The method according to claim 13, wherein forming the light-emitting unit comprises: providing a growth substrate; forming a fourth dielectric layer covering the growth substrate; etching the fourth dielectric layer to form at least one fourth groove, at least one fifth groove, and at least one sixth groove that expose the growth substrate, wherein the at least one fifth groove surround the at least one fourth groove, the at least one sixth groove surround the at least one fifth groove, and projected areas of the at least one fourth groove, the at least one fifth groove and the at least one sixth groove on the growth substrate are different; and epitaxially manufacturing the first light-emitting structure, the second light-emitting structure and the third light-emitting structure that are configured to emit light with different wavelengths in one epitaxial process in the at least one fourth groove, the at least one fifth groove and the at least one sixth groove respectively.

16. The semiconductor structure according to claim 1, wherein the first light-emitting structure the second light-emitting structure and the third light-emitting structure respectively comprise an N-type semiconductor layer, an active layer, and a P-type semiconductor layer that are stacked in sequence.

17. The semiconductor structure according to claim 16, wherein the N-type semiconductor layer is made of N-type doped Group III nitride-based materials, the P-type semiconductor layer is made of P-type doped Group III nitride-based materials, and group III nitride materials comprises any one or any combination of GaN, AlGa_N, InGa_N and AlInGa_N.

18. The semiconductor structure according to claim 1, wherein the first light-emitting structure, the second light-emitting structure and the third light-emitting structure are spaced apart from each other, space between the first light-emitting structure, the second light-emitting structure and the third light-emitting structure are filled with a dielectric material, and the dielectric material is made of silicon dioxide.

19. The semiconductor structure according to claim 11, wherein the first light-emitting structure, the second light-emitting structure and the third light-emitting structure respectively emit blue light, green light, and red light.

20. The semiconductor structure according to claim 12, wherein the light conversion layer is made of quantum dot or phosphor.
